CS1910 Computer Science I • File I/O **Partner workbook**

Turtle Instructions from a File

In lecture with a partner • about 15 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

Programs often read their instructions or data from a file. You will read a file of turtle commands, turn each line into a tuple, and work out what the commands draw.

1 Read the instructions

about 15 minutes

The problem

The file `instructions.txt` has one command per line: a word and a whole number. Read each line into a **tuple** containing the command and its integer value.

instructions.txt

```
FORWARD 100
RIGHT 90
FORWARD 100
RIGHT 90
FORWARD 100
RIGHT 90
FORWARD 100
RIGHT 90
```

read →

```
[('FORWARD', 100),
 ('RIGHT', 90),
 ('FORWARD', 100),
 ...
 ('RIGHT', 90)]
```

command integer value

1. What does one line of the file look like as a string, after it is read? What two steps turn it into a tuple?

2. Write code that reads the file into a list of tuples called `instructions`.

3. Follow the instructions by hand. What shape do they draw? Sketch it.

your sketch

4. Why is a tuple a good choice for each instruction?

✓ Before you leave

about 1 minutes

- [ ] I can open a text file and loop over its lines.
- [ ] I can split a line and convert its parts to the right types.
- [ ] I can store structured records as tuples in a list.